

18/10/22

Roll No.

TME-502

**B. TECH. (ME) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
DESIGN OF MACHINE ELEMENT-I**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain in detail the basic procedure with various steps involved in the design of machine element.
10 Marks (CO1/CO2)

OR

- (b) Briefly discuss BIS criteria for the specifications of plain carbon steel. Explain designation of (1) Fe320, (2) FeE200 and (3) 40C8.

10 Marks (CO1/CO2)

2. (a) A manufacturer is interested in manufacturing 5 different models of generator sets in the range of 5 kW to 50 kW capacity. Specify the capacities of the models. If he further wishes to expand this to 9 models within the same range, what will be the capacities of the additional models ?

10 Marks (CO1/CO2)

OR

- (b) How will you designate the steel with the following composition ?
C = 45 to 55%; Cr = 18%; Va = 21%

Explain designation – 40Cr13Mo10V2

10 Marks (CO1/CO2)

P. T. O.

3. (a) Explain the different modes of failure of machine element in detail.

10 Marks (CO2/CO3)

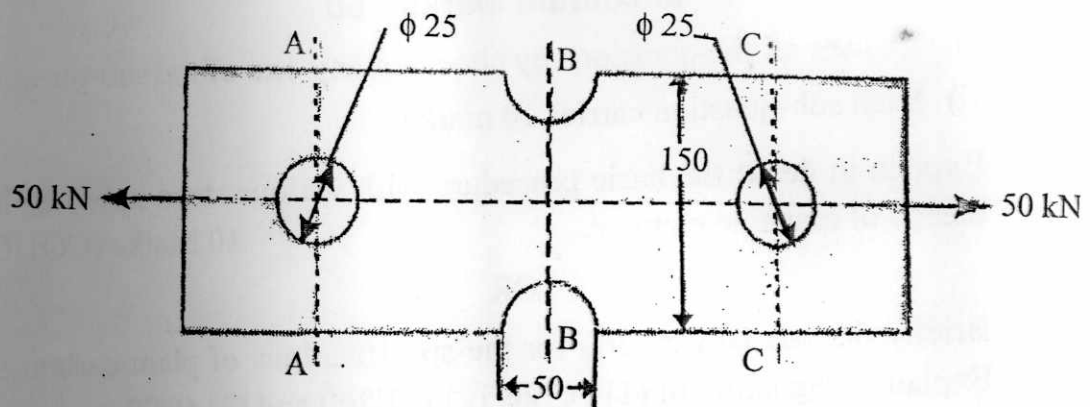
OR

- (b) How the factor of safety is evaluated for different types of loading ?
Specify the ranges for different cases.

10 Marks (CO2/CO3)

4. (a) A flat bar shown in the figure (Fig. 1) is subjected to an axial load of 50 kN. If the stress in the bar is to be limited to 100 N/m^2 , determine the thickness of the bar.

10 Marks (CO2/CO3)



OR

- (b) The stresses developed at a critical point in a machine component made of steel 45C8 ($S_{yt} = 380 \text{ MPa}$) are as follows :

10 Marks (CO2/CO3)

$$\sigma_x = 100 \text{ MPa}, \sigma_y = 40 \text{ MPa}, \tau_{xy} = 80 \text{ MPa}$$

Calculate the factor of safety by :

- Maximum principal stress theory
- Maximum shear stress theory
- Distortion energy theory

(3)

5. (a) Enumerate various theories of failure. Explain the criteria of failure for ductile materials and brittle materials subjected to static load.

10 Marks (CO2/CO3)

OR

- (b) Compare the various failure theories in terms of their safe zone profile for a specific machine component.

10 Marks (CO2/CO3)